COMPONENT SELECTION

Optical object-detection board

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Purpose and design status

This individual board will detect an opaque object interrupting an infrared beam and report object presence to the team system. The provisional arrangement places an emitter and detector across an approximately 50 mm opening indoors. A response time of 100 ms or less is a design target, not a measured result. This is presence detection; it does not measure distance.

Team coordination

The team requirements and block diagram have not yet been finalized. Optical object detection is my proposed individual subsystem. Mechanical spacing, object size, bus ownership and logic levels must be agreed with Team 103 before schematic release. The selections below are justified for this provisional scope and may change after that review.

Course constraints

Use the assigned PIC18F57Q43 Curiosity Nano, an individual 9 V barrel supply and onboard 5 V linear regulation. Build the peripheral circuitry from individual components on the PCB; do not use peripheral daughterboards. Include the required fuse and power indication in the schematic. The 2×4 IDC interface has five digital signals, two analog signals and ground; it does not distribute 5 V between boards.

Signal path

9 V adapter → local 5 V regulator → PIC and analog circuit. PIC output → PN2222 switch → 940 nm IR emitter → optical opening → BPW34 photodiode → MCP6002 transimpedance amplifier → PIC ADC → object-present output.

Comparison scope

Six major component groups are compared, with three candidates in each table. The microcontroller is assigned and is not re-selected. Ordinary resistors, capacitors, connectors, fuse, indicator LED, wiring and PCB are supporting items and are excluded from these major-component tables. The infrared emitter is included because it performs the sensing function. No motor, servo, solenoid or other major electromechanical actuator is proposed.

Custom versus ready-made

The selected implementation uses commercially available bare components in a custom receiver, driver and power circuit. A ready-made IR sensor breakout could reduce assembly effort, but would violate the course restriction on peripheral daughterboards and hide the required analog design. A phototransistor/resistor receiver is a simpler custom alternative; the chosen photodiode/amplifier approach gives explicit control of gain and analog filtering.

Pricing and source notes

Prices are USD as viewed on 19 October 2026, excluding tax, tariffs and shipping. Pack prices are normalized where indicated; confirm price and availability before ordering. Each row links its supplier listing and available datasheet. Product photos are supplier reference images, not photographs of parts already purchased or tested.

Course project requirements • Component Selection assignment

01 Photodetector

Optical object-detection subsystem — receiver

Three candidate solutions • supplier links and photos below

Part / unit cost	Advantages for this design	Limitations / tradeoffs
Vishay BPW34 $1.37  PIN photodiode; 7.5 mm² active area; 430–1100 nm spectral response; through-hole. Product & price Datasheet	• Large active area makes alignment across a short opening more forgiving. • Photocurrent can be converted with a chosen feedback resistance, giving direct control of analog gain.	• No internal gain: needs a low-bias-current amplifier and careful high-impedance layout. • Broad visible/IR response makes ambient-light rejection and saturation testing necessary.
Vishay TEFT4300 $0.70  Phototransistor; peak sensitivity 925 nm; daylight-blocking 3 mm package; ±30° half-angle. Product & price Datasheet	• Internal current gain can simplify the receiver to a resistor and ADC input. • Daylight-blocking package reduces visible-light interference with a 940 nm emitter.	• Transistor gain varies, making a repeatable calibrated analog transfer less direct. • Strong illumination can saturate the transistor and slow recovery; resistor selection still requires testing.
Everlight PT908-7C-F $0.95  940 nm phototransistor; side-looking through-hole package; SparkFun SEN-19018. Product & price Datasheet	• Side-looking geometry can place the sensing path parallel to the PCB. • Internal gain reduces the number of receiver components needed for a simple threshold detector.	• Gain and ambient-light response require threshold calibration for each assembled board. • Side-view mounting constrains the fixture orientation; saturation can limit response during large light changes.

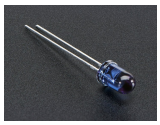
Selection: BPW34

I select the BPW34 because it supports a custom, adjustable analog receiver and provides a large target for alignment. The phototransistors can simplify detection, but their gain variability makes the analog response less predictable. I will use optical shielding and emitter-on/off sampling to address the BPW34's ambient-light sensitivity.

02 Infrared emitter

Optical object-detection subsystem — light source

Three candidate solutions • supplier links and photos below

Part / unit cost	Advantages for this design	Limitations / tradeoffs
Everlight IR333-A $0.75  940 nm IR LED; 5 mm through-hole; 20° beam width; Adafruit product 387. Product & price Datasheet	• Narrow beam concentrates light toward the opposite detector rather than nearby surfaces. • Standard 5 mm body is easy to mount in a drilled alignment fixture; bare LED meets the no-module rule.	• Narrow beam requires mechanical alignment and can be lost if the fixture moves. • Requires a current-limiting resistor and switch; invisible emission cannot be checked by eye.
Vishay TSAL6200 $0.50  940 nm IR LED; 5 mm package; 34° full viewing angle; 100 mA maximum continuous rating. Product & price Datasheet	• Lowest listed emitter price and a conventional through-hole footprint. • Higher current capability leaves room for optical-margin experiments after thermal and drive checks.	• Wider beam than IR333-A can illuminate surrounding surfaces and reduce beam-break contrast. • The 100 mA rating is not the proposed operating current; using it would increase power and transistor-drive needs.
Everlight IR908-7C-F $0.95  940 nm IR LED; side-looking package; 60° viewing angle; SparkFun SEN-18772. Product & price Datasheet	• Side-looking shape suits a beam traveling along the surface of a PCB. • Wider emission angle relaxes initial alignment and matches the PT908 receiver package style.	• Wide emission spreads optical power and increases unwanted reflections near the sensing opening. • Higher price than the other two options; a fixture must support its side-view geometry.

Selection: IR333-A

I select IR333-A for its narrower beam and familiar 5 mm mounting. The $0.25 premium over TSAL6200 is small relative to the benefit of directing light across the opening. Its alignment requirement will be handled with a fixed holder. Proposed operation is approximately 20 mA, subject to bench verification.

03 Analog amplifier

Optical object-detection subsystem — signal conditioning

Three candidate solutions • supplier links and photos below

Part / unit cost	Advantages for this design	Limitations / tradeoffs
Microchip MCP6002-I/P $0.45  Dual rail-to-rail input/output op amp; 1.8–6 V supply; 1 MHz gain-bandwidth; PDIP-8. Product & price Datasheet	• Very low typical input bias current is suitable for a photodiode transimpedance stage. • One IC supplies both the signal amplifier and a buffered midpoint reference; rail-to-rail operation fits 5 V.	• Low-current, 1 MHz design limits high-speed operation and direct heavy-load drive. • Offset, photodiode capacitance and PCB leakage still require error/stability checks; rail-to-rail does not mean exact rail swing under load.
Texas Instruments LM358P $0.33  Dual op amp; 3–30 V supply; about 0.7 MHz gain-bandwidth; PDIP-8. Product & price Datasheet	• Lowest listed amplifier price and an easy-to-solder dual through-hole package. • Works from the required single 5 V rail, with input common-mode range extending to ground.	• Input/output do not extend to the positive rail, reducing the usable signal range on 5 V. • Higher input bias current than the CMOS options produces a larger error through the feedback resistor.
Texas Instruments TLV2372IP $2.38  Dual rail-to-rail input/output op amp; 2.7–16 V supply; 3 MHz gain-bandwidth; PDIP-8. Product & price Datasheet	• Greater bandwidth provides more settling-time margin if emitter modulation is increased. • CMOS input and rail-to-rail operation suit the same photodiode/5 V architecture.	• Costs over five times the MCP6002 at the listed single-unit prices. • About 750 µA typical supply current per channel is higher, while the initial slow beam detector does not need the extra bandwidth.

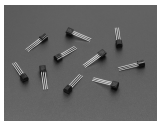
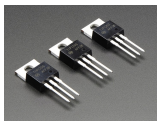
Selection: MCP6002-I/P

I select MCP6002-I/P for low input bias current, 5 V rail-to-rail operation and two channels in one DIP package. It fits a custom transimpedance amplifier plus reference buffer. LM358 sacrifices signal range; TLV2372 adds speed and cost that the initial response target does not require.

04 Emitter switching transistor

Optical object-detection subsystem — emitter drive

Three candidate solutions • supplier links and photos below

Part / unit cost	Advantages for this design	Limitations / tradeoffs
PN2222 NPN transistor $0.195 each[1]  Small-signal NPN, through-hole TO-92; Adafruit product 756, pack of 10. Product & price Datasheet	• Inexpensive low-side switch with ample current capability for a proposed 20 mA emitter. • Resistor-limited base drive provides a straightforward way to pulse the emitter from the PIC.	• Needs continuous base current while on, unlike a MOSFET. • Saturation voltage reduces LED-resistor headroom; verify the supplied part's pinout before making the footprint.
onsemi 2N7000 $0.66  N-channel MOSFET; TO-92; 60 V, 200 mA maximum ratings. Product & price Datasheet	• Very small steady-state gate current reduces load on the PIC output. • Through-hole package is simple to assemble and can replace the low-side switching function.	• Gate threshold is not a guarantee of low on-resistance; verify behavior at the actual GPIO drive voltage. • Gate is sensitive to ESD and needs a defined off state; costs more than a PN2222 from the pack.
TIP120 Darlington transistor $0.833 each[2]  NPN Darlington power transistor; TO-220; Adafruit product 976, pack of 3. Product & price Datasheet	• High current gain reduces base-drive requirements relative to an ordinary power BJT. • Large package is easy to handle and could support substantially larger loads in a different application.	• Darlington voltage loss wastes headroom and power in this low-current 5 V emitter circuit. • TO-220 body occupies much more area than needed for 20 mA; higher cost and unnecessary current capacity.

Selection: PN2222

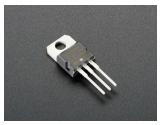
I select PN2222 for the approximately 20 mA LED load. A proposed 2 mA base-drive allowance gives a forced gain of 10; the resistor will be selected using the actual PIC output voltage and transistor data. A base pull-down will keep the emitter off during reset. The MOSFET is viable, but requires its on-resistance to be checked at the chosen logic voltage.

[1] $1.95 / 10 pieces. [2] $2.50 / 3 pieces. Actual purchase cost is the full pack price.

05 5 V linear regulator

Optical object-detection subsystem — local power regulation

Three candidate solutions • supplier links and photos below

Part / unit cost	Advantages for this design	Limitations / tradeoffs
ST L7805CV $0.75  Fixed 5 V linear regulator; TO-220; up to 1.5 A with adequate thermal design. Product & price Datasheet	• Directly supplies the required 5 V rail from the specified 9 V adapter. • Fixed output avoids an adjustment-divider error; simple support circuit and low listed price.	• Approximately 4 V is dissipated as heat at nominal input; current rating alone does not establish safe loading. • Needs input headroom and local bypassing; thermal rise must be checked in the actual enclosure.
onsemi LM317TG $1.95  Adjustable linear regulator; TO-220; nominal 1.2–37 V output range. Product & price Datasheet	• A resistor divider can set the required 5 V output and permit later voltage changes. • Through-hole package and current/thermal limiting support straightforward prototyping.	• Requires additional setting resistors and minimum-load checks for a rail that is fixed at 5 V. • Still dissipates the 9-to-5 V drop; costs more and adds adjustment tolerances without a needed benefit.
TI LM2940CT-5.0/NOPB $2.04  Fixed 5 V, 1 A low-dropout linear regulator; TO-220; approximately 0.5 V typical dropout at 1 A. Product & price Datasheet Photo source	• Lower dropout preserves regulation if input voltage falls close to 5 V. • Includes protective functions and a convenient through-hole package.	• Dropout advantage gives little benefit with a regulated 9 V input; linear conversion still produces heat. • Requires at least 22 µF output capacitance with a suitable ESR and is more expensive than L7805CV.

Selection: L7805CV

I select L7805CV because the course specifies a 9 V input and an onboard 5 V linear regulator. It is the least expensive fixed-output choice here. The preliminary power check predicts noticeable heating, so final acceptance depends on measured load and a thermal check, with a heatsink if necessary.

06 9 V power source

Optical object-detection subsystem — external power

Three candidate solutions • supplier links and photos below

Part / unit cost	Advantages for this design	Limitations / tradeoffs
Adafruit 9 V, 1 A adapter $8.95  Regulated 9 V DC; center-positive 5.5 × 2.1 mm barrel connector; product 63. Product & price Datasheet	• Matches the proposed barrel-jack voltage, polarity and size. • 1 A capacity comfortably exceeds the preliminary single-board allocation; lowest price of these adapters.	• Tethered mains operation does not provide portable battery use. • Connector polarity must be checked; a 1 A source does not remove the linear regulator’s thermal limit.
Pololu 9 V, 1 A adapter $11.11  Regulated 9 V DC; center-positive 5.5 × 2.1 mm barrel connector; product 1470. Product & price	• Same electrical output and connector format provide a practical alternate supplier. • Sufficient current reserve for the preliminary board allocation without an oversized desktop supply.	• Higher listed price than Adafruit for the same nominal current capacity. • Still requires an outlet and provides no improvement to 7805 heat dissipation.
Pololu 9 V, 5 A adapter $36.80  Regulated 9 V DC; center-positive 5.5 × 2.1 mm barrel connector; product 1465. Product & price	• Large current reserve could serve a substantially higher-power load in another design. • Same nominal voltage and connector format simplify electrical compatibility checks.	• Cost and physical bulk are disproportionate to this sensing board’s current needs. • Extra available fault current makes appropriate branch protection important; it cannot increase the safe current of the 5 V regulator.

Selection: Adafruit product 63

I select the Adafruit regulated 9 V, 1 A adapter. It supplies ample capacity for the preliminary allocation and has the lowest listed adapter price. Pololu 1470 is a reasonable alternate source. The 5 A option adds cost and bulk without a benefit for this individual sensing board.

Implementation plan

Initial analog circuit

Use one MCP6002 channel to buffer a 2.5 V midpoint from a 10 kΩ/10 kΩ divider. Use the second channel as a transimpedance amplifier: non-inverting input at 2.5 V, BPW34 cathode at the inverting summing node, anode at ground, and feedback resistor/capacitor from output to the summing node. With this polarity, increasing photocurrent raises the output.

Starting values, not final specifications

A starting feedback choice is 100 kΩ in parallel with 1 nF. The ideal feedback pole is $1/(2\pi RC) \approx 1.59$ kHz. An assumed 0–20 µA photocurrent range gives approximately 2.5–4.5 V output. This current range is a sizing assumption, not a prediction at 50 mm. Select gain after measuring clear-beam and blocked-beam currents. Verify stability with photodiode and board capacitance.

Detection method

Start with 50 Hz emitter on/off cycles. After settling, measure the ADC in both states; subtract the off sample from the on sample to reduce slowly varying ambient light. Compare the difference with a calibrated threshold and hysteresis. A blocked beam should give a smaller difference. Test under room lighting and near windows; ambient saturation cannot be corrected by subtraction alone.

Construction and verification

Keep the photodiode/summing-node traces short and clean; place decoupling next to the amplifier and regulator. Keep switched LED current away from the sensitive analog return. Verify the supplied transistor lead order and every footprint. Check regulator temperature, ADC range and PIC external-power instructions before connecting USB and external power together. No circuit has been built or tested for this selection report.

Selected major-part cost

Selected part	Qty used	Cost used
BPW34	1	$1.37
IR333-A	1	$0.75
MCP6002-I/P	1	$0.45
PN2222	1	$0.195
L7805CV	1	$0.75
9 V / 1 A adapter	1	$8.95
Selected major parts total		≈ $12.47

Using the full PN2222 pack, the initial purchase subtotal for these six selections is $14.22. This is not a complete board BOM: MCU/board, PCB, passives, connector, fuse, fixture, possible heatsink and shipping/taxes are excluded.

Proposed interface & review

Connection	Proposed signal	Direction / state	Purpose
PIC RA0	Receiver ADC	Analog input	Read conditioned photodiode voltage
PIC RA1	Emitter control	Digital output; reset OFF	Drive PN2222 base resistor
IDC 1 / PIC RB0	OBJECT_PRESENT	Digital output after team approval	Report blocked beam
IDC 2 / PIC RB1	SENSOR_ENABLE	Digital input	Optional team enable; default defined locally
IDC 3–5	Digital reserved	Undriven / high impedance	Assign with team before routing
IDC 6–7	Analog reserved	Undriven / high impedance	No analog bus output proposed
IDC 8	Common ground	Ground	Shared signal reference
Local barrel jack	9 V input	Power input	Center positive; onboard 5 V conversion

Port names are proposed logical assignments, not verified Curiosity Nano header positions. Confirm the board guide, ADC/PPS setup, available pins, board supply configuration and team logic levels before schematic capture. Keep all bus outputs disabled until ownership is agreed; do not connect two push-pull outputs together.

Before the next design milestone

• Agree the team function, optical opening, object size and interface ownership.

• Confirm Curiosity Nano external-power wiring and all proposed pin assignments.

• Measure clear/blocked optical levels; adjust feedback gain, LED current and hysteresis.

• Check response time, ambient immunity and regulator temperature on the assembled prototype.

• Complete the course power-budget worksheet with documented worst-case MCU/board current and all supporting loads.

Supporting page

The separate Power Budget page records preliminary allocations and the remaining maximum-current verification. Power budgeting is optional for the initial Component Selection submission but must be completed for external design review.

Assignment requirements